Lecture 1 (September 2nd)

Remark. The main subject of this course is Galois theory. The motivation of this theory is to find a general solution of polynomial equations. For example, the general solution of the quadratic equation

$$ax^2 + bx + c \;=\; 0$$

where $a \neq 0$, b and c are real numbers is given by

$$x \;=\; \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

To be more precise, we want to express a general solution of the equation

$$a_n x^n + a_{n-1} x^{n-1} + \ldots + a_0 = 0$$

where $a_i \in \boldsymbol{Q}$ and $a_n \neq 0$ using 4 basic operations (namely, $\pm$, $\times$, and $\div$), and n-th roots. Today, we know that, through Galois theory, this is not possible for $n \geq 5$.

In essence, Galois theory connects the theory of fields and the theory of groups. Consider the following example. The polynomial

$$x^4 + x^3 + x^2 + x + 1 = 0$$

We know how to solve this,

$$(x-1)(x^4 + x^3 + x^2 + x + 1) = x^5 - 1 = 0$$

and thus we choose a primitive 5th root of unity (say $\xi = \exp(2\pi i/5)$). Then the solutions are

$$(\xi^1, \xi^2, \xi^3, \xi^4)$$

We see that this group is homomorphic to $\boldsymbol{Z}/5\boldsymbol{Z}$, and that there's a certain relationship between the roots of this polynomial and a certain group.

Chapter 13. Field Extensions

Definition. (13.1) Let K be a field. A field extension of K is another field F endowed with an injective ring homomorphism $\phi : K \to F$. You are essentially identifying the field with a subfield of a larger ring F.

Remark. From now on, we use the following conventions:

(i) Every ring is a commutative ring with 1

(ii) A ring homomorphism sends 1 to 1

Remark. There is at most one ring homomorphism between two fields. Notice that $\ker \phi$ is an ideal (additive subgroup with the absorption property) of K, so that $\ker \phi$ is either 0 or K. Therefore, ϕ must be injective.

Lecture 2 (September 4th)

Definition. (13.1) Let K be a field. A field extension of K is a pair $(F, i : K \to F)$ where F is a field and i is an injective (ring) homomorphism. We draw field extension tower diagrams like the following:

$$\begin{tikzcd} F \arrow[d, dash] \\ K \end{tikzcd}$$

Example. (13.2 $*$) (Algebraic extension) Let $f(t) \in \boldsymbol{Q}[t]$ be a nonzero irreducible polynomial. Then $\boldsymbol{Q}[t] \,/\, (f(t))$ is a field, and the composite map

$$\boldsymbol{Q} \to \boldsymbol{Q}[t] \to \boldsymbol{Q}[t] \,/\, (f(t))$$

is a field extension. This is the general way we would be obtaining field extensions.

$$\begin{tikzcd} \boldsymbol{Q}[t] \,/\, (f(t)) \arrow[d, dash] \\ \boldsymbol{Q} \end{tikzcd}$$

Proof. The set $\boldsymbol{Q}[t] \,/\, (f(t))$ is a maximal ideal and since $f(t)$ is a nonzero irreducible element of $\boldsymbol{Q}[t]$ which is principle ideal domain. Therefore, the quotient $\boldsymbol{Q}[t] \,/\, (f(t))$ is a field. □

Example. (13.3) (Transcendental extension) Let $\boldsymbol{Q}(t)$ denote the field of fractions of $\boldsymbol{Q}[t]$. Then the composition map $\boldsymbol{Q} \to \boldsymbol{Q}[t] \to \boldsymbol{Q}(t)$ determines a field extension.

Definition. (13.6) The characteristic of a field K ($\mathrm{char}(K)$) is the smallest nonnegative generator of the kernel of the unique ring homomorphism $\phi : \boldsymbol{Z} \to K$.

Lemma. (13.7) Let $F \,/\, K$ be a field extension. Then

$$\mathrm{char}(K) = \mathrm{char}(F)$$

Lemma. (13.8) Let F be a field. Then F contains a unqiue copy of $\boldsymbol{Q}$ or $\boldsymbol{Z} \,/\, p\boldsymbol{Z}$, where p is prime.

13.2 Finite Extensions and the Degree of an Extension

Definition. (13.10) The degree of a field extension $F \,/\, K$, denoted $[F : K]$, is the dimension of F as a vector space over K.

Remark. If $i : K \to F$ is a field extension, then we can view F as a K-vector space through the map i.

Definition. (13.11) A field extension is finite if $[F : K] < \infty$.

Definition. Let F/K be a field extension, and let $\alpha \in F$. We will say that α is algebraic over K if there exists a nonzero polynomial $f(t) \in K[t]$ such that $f(\alpha) = 0$. The element $\alpha \in F$ is transcendental if not algebraic.

Example. Consider

$$\begin{array}{c} \boldsymbol{C} \\ | \\ \boldsymbol{R} \end{array}$$

(i) Is i algebraic over $\boldsymbol{R}$? Yes! i is a zero of $t^2 + 1 \in \boldsymbol{R}[t]$.

(ii) $\sqrt{2}$ is algebraic over $\boldsymbol{Q}$

(iii) Also, $\pi, e \in \boldsymbol{C}$ is transcendental over $\boldsymbol{C}$.

(iv) Every element α of K is algebraic over K $(t - \alpha \in K[t])$.

Definition. (13.20) Let $F \,/\, K$ be a field extension. If $\alpha \in F$ is algebraic, the minimal polynomial of α over K is the unqiue monic irreducible polynomial $p(t) \in K[t]$ such that $f(\alpha) = 0$. To understand this, consider a ring homorphism

$$\underset{\alpha}{\mathrm{ev}} : K[t] \to F$$

given by evaluation at α (namely, $f(t) \mapsto f(\alpha)$). Note that $\ker(\mathrm{ev}_\alpha)$ is something because α is algebraic in K.

Lecture 3 (September 9th)

Remark. Last time, we learned about algebraic elements. This class, we learn about algebraic extensions.

Definition. (13.11) A field extension $F \,/\, K$ is finite if $[F : K]$ is finite.

Definition. $\alpha \in F$ is algebraic over K if $p(\alpha) = 0$ for some nonzero $p(t) \in K[t]$. Otherwise, α is called transcendental over K.

Example. (i) $\alpha \in K \subset F$ is algebraic over K

(ii) $i \in \boldsymbol{C}$ is algebraic over $\boldsymbol{R}$

(iii) $\sqrt{3} \in \boldsymbol{R}$ is algebraic over $\boldsymbol{Q}$

Definition. (13.20) If $\alpha \in F$ is algebraic over K, the minimal polynomial of α over K is the unique monic irreducible polynomial $p(t) \in K[t]$ such that $p(\alpha) = 0$. To understand this definition, consider the evaluation homomorphism

$$\mathrm{ev}_\alpha : K[t] \to F : f(t) \mapsto f(\alpha)$$

where

$$\sum_{i=0}^{n} \alpha_i t^i \mapsto \sum_{i=0}^{n} a_i \alpha^i$$

(1) Then its kernal $\ker \mathrm{ev}_\alpha$ is of the form $(p(t))$ for some $p(t) \in K[t]$ (because $K[t]$ is a principle ideal domain).

(2) By the 1st isomorphism theorem, it follows that

$$K[t] \,/\, \ker \mathrm{ev}_\alpha = K[t] \,/\, (p(t))$$

is isomorphic to $\mathrm{im}\, \mathrm{ev}_\alpha$.

(3) $\mathrm{im}\, \mathrm{ev}_\alpha = K[\alpha]$

(4) We claim that $p(t) \in K[t]$ is irreducible. Note that

$$K[t] \,/\, (p(t)) \cong \mathrm{im}\, \mathrm{ev}_\alpha = K[\alpha] \subset F$$

is an integral domain (since F is a field and $K[\alpha]$ is its subring). This implies that $(p(t))$ is a prime ideal of $K[t]$. Using this fact, and the fact that α is algebraic, we conclude that $p(t) \in K[t]$ is irreducible.

(5) We may suppose that $p(t)$ is monic.

(6) So far, we have a monic irreducible $p(t) \in K[t]$ with $p(\alpha) = 0$. The uniqueness of $p(t)$ is still required. Suppose $q(t) \in K[t]$ is a monic irreducible polynomial such that $q(\alpha) = 0$. Then $q(t) \in \ker \mathrm{ev}_\alpha$, so that $q(t) = r(t) \cdot p(t)$ for some $r(t) \in K[t]$. Since $q(t) \in K[r]$ is irreducible and $p(t)$ is a nonunit,

$$r(t) \in (K[t])^\times = K^\times$$

Write $r(t) = r$. We deduce that $r = 1$ (since both $p(t)$ and $q(t)$ are monic).

Definition. We let $K[\alpha]$ denote the subring generated by K and α. We let $K(\alpha)$ denote the subfield generated by K and α.

Remark. Take a principle ideal domain (PID) R and a nonzero $p \in R$. The following are equivalent

(i) p is prime

(ii) $(p) \subset R$ is prime

(iii) $(p) \subset R$ is maximal

(iv) p is irreducible

Remark. We define the following

$$K[\alpha] = \{f(\alpha) \in F \mid f(t) \in K[t]\}$$

and

$$K(\alpha) = \left\{\frac{f(\alpha)}{g(\alpha)} \in F \mid f(t), g(t) \in K[t] \ g(\alpha) \neq 0\right\}$$

Lecture 5 (September 16th)

Recall. Last time, we have dealt with $K[x] \,/\, (p(t))$ and minimal polynomials. Today, we will learn about properties of finite and algebraic extensions.

Definition. (Simple field extensions) A field extension $F \,/\, K$ is simple if $F = K(\alpha)$ for some $\alpha \in F$, where $K(\alpha)$ denotes the smallest subfield of F containing $K \cup \{\alpha\}$.

Definition. (13.20) (Minimal polynomial) If $\alpha \in F$ is algebraic over K, the minimal polynomial of α over K is the unique monic irreducible polynomial $p(t) \in K[t]$ with $p(\alpha) = 0$.

Remark. (Notation) $K(\alpha)$ is the smallest subfield, and $K[\alpha]$ is the smallest subring of F containing $K \cup \{\alpha\}$.

Remark. (i) $\alpha \in F$ is algebraic over K if and only if

$$\ker\Big(\mathrm{ev}_\alpha : K[t] \to F : f(t) \mapsto f(\alpha)\Big)$$

is nontrivial.

(ii) If α is algebraic over K, then $K[\alpha] = K(\alpha)$. In particular, every element of $K(\alpha)$ can be expressed as a polynomial in α with coefficient in K.

Proof. Consider $\alpha \in F$.

$$\ker \mathrm{ev}_\alpha = (p(t))$$

and due to the first isomorphic theorem,

$$K[t] \,/\, (p(t)) \cong K[\alpha]$$

If α is algebraic over K, then $(p(t))$ must be a maximal ideal of $K[t]$. This implies that $K[\alpha] = K(\alpha)$. □

Proposition. (13.17) Read this!

Example. (13.21)

$$\alpha = 2^{1/5} \in \boldsymbol{Q}(2^{1/5})$$

Notice that π is algebraic in $\boldsymbol{R}$ but is not algebraic in $\boldsymbol{Q}$.

Theorem. (7.11) Let K be a field and $f(x)$ be a nonzero polynomial of degree n. Then,

$$K[x] \,/\, (f(x))$$

is a n-dimensional vector space over K.

Proof. The following set is a basis for $K[t] \,/\, (f(t))$

$$\{1, x, \ldots, x^{n-1}\}$$

□

Remark. (Notation) The minimal polynomial of α over $K = \mathrm{irr}(\alpha, K)$.

13.4 Algebraic Extensions

Definition. (13.30) A field extension $F \,/\, K$ is algebraic if every element of F is algebraic over K.

Proposition. (13.31) Every finite extension $F \,/\, K$ is algebraic.

Proof. Fix an element $\alpha \in F$. Set $\dim F = n$. Then

$$\{1, \alpha, \alpha^2, \ldots, \alpha^n\}$$

is linearly dependent over K, that is, $a_0 + a_1\alpha + \ldots + a_n\alpha^n = 0$ for some $(a_0, \ldots, a_n) \neq 0 \in K^{n+1}$. Take

$$f(t) = a_0 + a_1 t + \ldots + a_n t^n$$

□